Calcurion Maths Worksheets

A-Level Mathematics

Topic: Pure 1 – Circles

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Find the centre and radius of $x^2 + y^2 = 25$. **[2]**

2. Find the centre and radius of $x^2 + y^2 - 4x + 6y = 0$. **[4]**

3. Find the centre and radius of $x^2 + y^2 + 2x - 8y - 8 = 0$. **[4]**

4. Find the centre and radius of $2x^2 + 2y^2 - 4x + 12y = 0$. **[4]**

5. Write the equation of the circle with centre $(3, -2)$ and radius 4. **[2]**

6. Write the equation of the circle with centre $(-1, 5)$ and radius $\sqrt{7}$. **[2]**

7. Does the point $(3, 4)$ lie on $x^2 + y^2 = 25$? **[2]**

8. Does the point $(1, 2)$ lie on $x^2 + y^2 - 2x - 4y = 0$? **[3]**

9. Find the equation of the circle with diameter endpoints $A(1, 2)$ and $B(5, 6)$. **[4]**

10. Find the equation of the circle with diameter endpoints $A(-1, 3)$ and $B(3, 1)$. **[4]**

11. Find the points of intersection of the line $y = x + 1$ and the circle $x^2 + y^2 = 25$. **[5]**

12. Find the points of intersection of $y = 2x$ and $x^2 + y^2 = 20$. **[4]**

13. Show that the line $y = 3x + 1$ does not intersect $x^2 + y^2 = 5$. **[4]**

14. Find the equation of the tangent to $x^2 + y^2 = 25$ at $(3, 4)$. **[4]**

15. Find the equation of the tangent to $x^2 + y^2 = 13$ at $(2, 3)$. **[4]**

Section B – Hard (Q16–25)

16. Find the equation of the tangent to $x^2 + y^2 - 4x + 2y = 0$ at $(1, 1)$. **[5]**

17. Find the length of the tangent from $(6, 0)$ to the circle $x^2 + y^2 = 16$. **[4]**

18. Find the length of the tangent from $(5, 5)$ to $x^2 + y^2 - 2x - 4y = 0$. **[5]**

19. The circle $x^2 + y^2 + ax + by + c = 0$ passes through $(1, 2)$ and $(3, 4)$ and has centre on $x = 2$. Find a, b, c. **[6]**

20. Find the points of intersection of the circles $x^2 + y^2 = 25$ and $x^2 + y^2 - 6x - 8y = 0$. **[6]**

21. Show that the circles $x^2 + y^2 - 4x = 0$ and $x^2 + y^2 - 6y = 0$ touch, and find the point of contact. **[6]**

22. Find the equation of the circle passing through $A(1,0)$, $B(0,1)$ and $C(0,-1)$. **[6]**

23. Find the equation of the normal to $x^2+y^2=25$ at $(3,4)$. **[4]**

24. The line $y=mx+c$ is tangent to $x^2+y^2=r^2$. Show that $c^2=r^2(1+m^2)$. **[5]**

25. Find the condition on k such that $x^2+y^2+2x-4y+k=0$ represents a real circle. **[4]**

End of Worksheet • *Show all working clearly.*